

GC University Lahore

BS Computer Science

Examination: Final 2025

Semester: 7th

COURSE CODE: CS-3216

TITLE: Parallel and Distributed Computing

Time allowed: 155 minutes

Marks: 40

Name: _____

Roll No: _____

Subjective

Attempt any Four Questions

Question No. 02:

10

A team is implementing a parallel sorting algorithm on a distributed memory system. High communication delays between nodes are significantly slowing down the sorting process. What measures can they take to minimize these delays and enhance sorting performance?

Question No. 03

10

What parameters are used to evaluate a static interconnection? Derive the generalized equation for all parameters.

Question No. 04

10

Draw a task independence graph for the following table having information about used cars.

ID#	Model	Year	Color	Dealer	Price
4523	Civic	2002	Blue	MN	\$18,000
3476	Corolla	1999	White	IL	\$15,000
7623	Camry	2001	Green	NY	\$21,000
9834	Prius	2001	Green	CA	\$18,000
6734	Civic	2001	White	OR	\$17,000
5342	Altima	2001	Green	FL	\$19,000
3845	Maxima	2001	Blue	NY	\$22,000
8354	Accord	2000	Green	VT	\$18,000
4395	Civic	2001	Red	CA	\$17,000
7352	Civic	2002	Red	WA	\$18,000

Question No. 05:

10

A university is teaching parallel programming using shared memory systems. Students often face data inconsistency issues when running multi-threaded programs. What techniques can they implement to ensure proper synchronization between threads?

Question No. 06:

10

A distributed system is used to process real-time stock market data. The system frequently synchronizes between nodes causing delays during high transaction volumes. What strategies can the team employ to reduce synchronization overhead while maintaining data consistency?